A global perspective on the social structure of science

Aliakbar Akbaritabar

Max Planck Institute for Demographic Research, Rostock, Germany

Akbaritabar@demogr.mpg.de

Andres F. Castro Torres

Center for Demographic Studies, Autonomous University of Barcelona, Barcelona, Spain

Vincent Larivière

Université de Montréal, Montréal, Canada.

Working paper

SCAN ME

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Problem

There are [*bibliometric*] myths about academia on **productivity**, **internationalization**, **mobility**, and **impact**.

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There are [*bibliometric*] myths about academia on **productivity**, **internationalization**, **mobility**, and **impact**.

Number of publications has increased a lot. **Collaboration** is the norm (*team science*).

> **33%** of 8 million authors in Scopus have **1** paper. Median N. of authors in 28+ million pubs: **2**, 75% quantile is 4 (**27.2%** solo authored).

High and increased share of **internationalization**

> **75%** of pubs are by authors from one country (national pubs).

Vast and growing number of **mobility** and it positively affects scholars' performance, collaborations and **impact**.

> **87.5%**, one country of affiliation (no international mobility), **73.5%**, one region (no sub-national mobility)

Gap

These processes need to be studied **simultaneously**, in an **integrated** framework and on a **global** scale.

To show that they **might not be as prevalent as the literature states** or the trends differ when the four processes are considered **together**.

Question and Method

Are authors who **collaborate the most,** also the **most mobile**, **productive**, and **cited**?

With whom do they collaborate? **Within** their group?

We selected 12 most widely used variables:

4 measuring **collaboration**

3 measuring **mobility**

3 measuring **productivity**

2 measuring **impact**

Multiple Correspondence Analysis

CLUSTER ANALYSIS

Co-authorship networks

Community detection

12 mostly used bibliometric variables

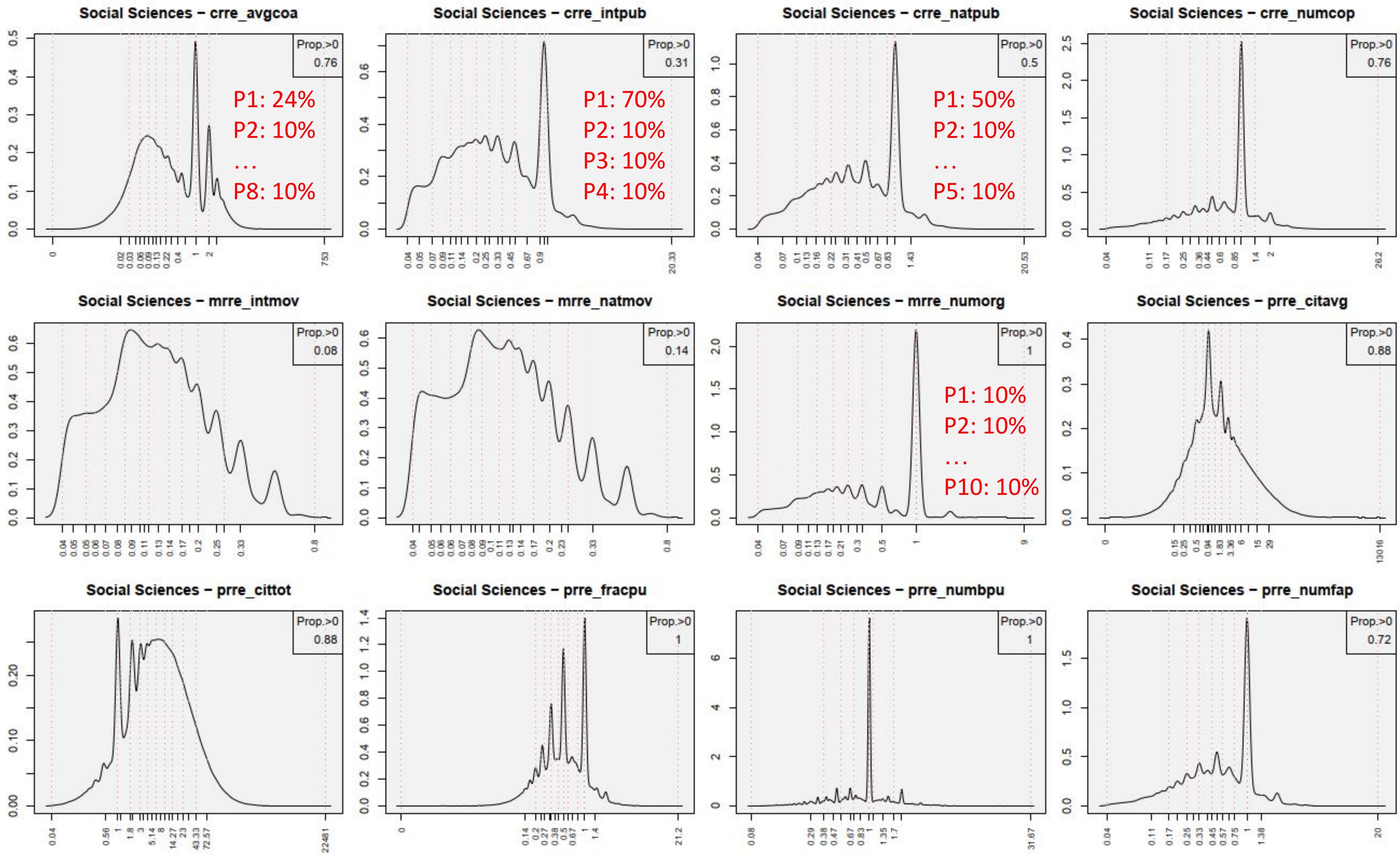
Productivity	**Number** of publications **Fractional count** of publications Number of **first-author** publications
Collaboration/ internationalization	Number of **coauthored** papers Average number of **coauthors** per paper Number of **internationally** coauthored publications Number of **nationally** coauthored publications
Mobility	Number of **international changes** in academic affiliation Number of **national changes** in academic affiliation Number of affiliated **organizations**
Impact/visibility	Average number of **citations** per paper **Total** number of citations

Data

Variable	Scopus*, 1996-2020
N. of publications (article/reviews)	28,461,324
N. of **authors** (with SCP author ID)	**8,225,368**
N. of authors (with SCP ID) with 1 publication (**33**%)	2,709,928
N. of institutions	53,787
N. of authors with 1 institution (**64.8**%)	5,327,397
N. of countries	205
N. of authors with 1 country (**87.5**%)	7,200,994
N. of regions	1,836
N. of authors with 1 region (**73.5**%)	6,042,077
N. of unique years	25
N. of authors with 1 year (**36.8**%)	3,025,586
N. of papers with 1 author (**27.2**%)	7,743,400

OECD FOS (+70% pubs, most category)	Count of authors
Agricultural Sciences	237,960
Engineering and Technology	786,541
Humanities	113,054
Medical and Health Sciences	**3,122,108**
Natural Sciences	**3,475,981**
No_discipline_assigned	41,278
Social Sciences	448,446

* Data from German Competence network for bibliometrics via MPDL



Social Sciences – crre_avgcoa
Prop.>0
0.76
P1: 24%
P2: 10%
…
P8: 10%
Social Sciences – crre_intpub
Prop.>0
0.31
P1: 70%
P2: 10%
P3: 10%
P4: 10%
Social Sciences – crre_natpub
Prop.>0
0.5
P1: 50%
P2: 10%
…
P5: 10%
Social Sciences – crre_numcop
Prop.>0
0.76
Social Sciences – mrre_intmov
Prop.>0
0.08
Social Sciences – mrre_natmov
Prop.>0
0.14
Social Sciences – mrre_numorg
Prop.>0
1
P1: 10%
P2: 10%
…
P10: 10%
Social Sciences – prre_citavg
Prop.>0
0.88
Social Sciences – prre_cittot
Prop.>0
0.88
Social Sciences – prre_fracpu
Prop.>0
1
Social Sciences – prre_numbpu
Prop.>0
1
Social Sciences – prre_numfap
Prop.>0
0.72

What if we combine these 12 variables to three inter-related axes with MCA?

Contributions

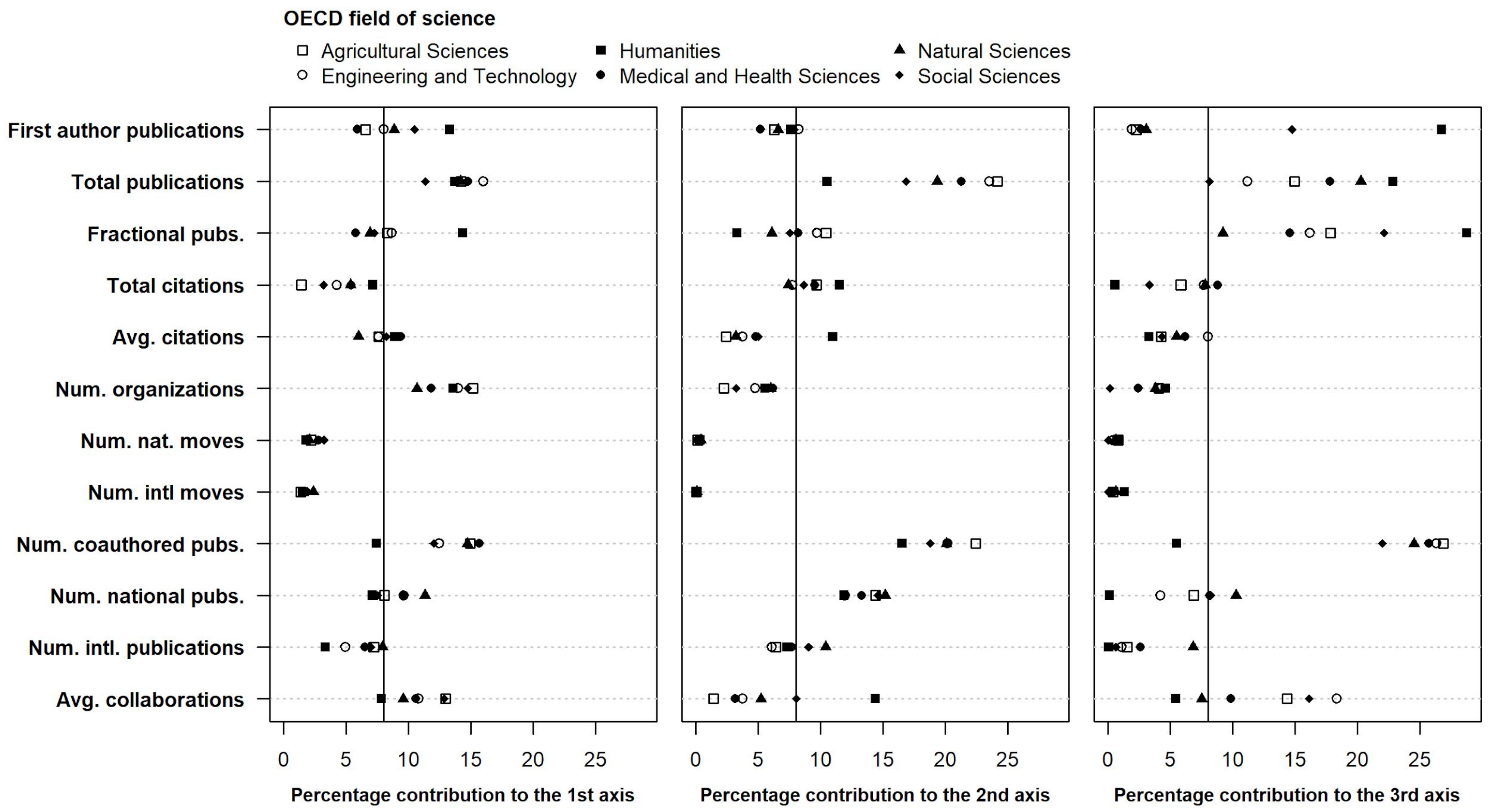
(i) include **extreme values** in the analysis,

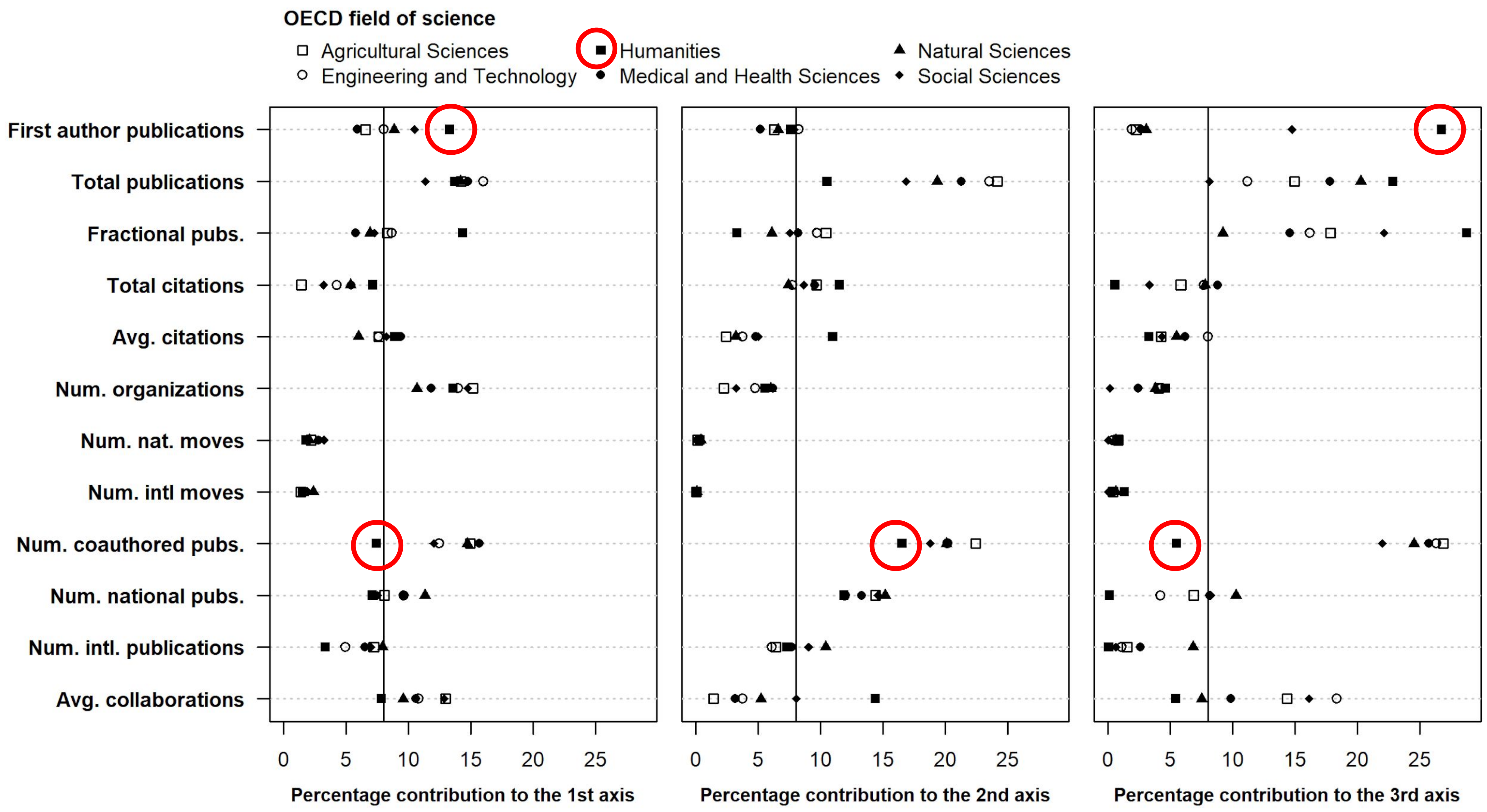
(ii) capture potential **non-linear relationships** among variables,

(iii) preserve the **distributional characteristics** of each indicator, and

(iv) avoids potential biases in correlational analyses due to **outlier observations**.

A **multidimensional measure** of **social stratification** within scientific communities


OECD field of science
Agricultural Sciences
Engineering and Technology
Humanities
Medical and Health Sciences
Natural Sciences
Social Sciences
First author publications
Total publications
Fractional pubs.
Total citations
Avg. citations
Num. organizations
Num. nat. moves
Num. intl moves
Num. coauthored pubs.
Num. national pubs.
Num. intl. publications
Avg. collaborations
0
5
10
15
20
25
Percentage contribution to the 1st axis
Percentage contribution to the 2nd axis
Percentage contribution to the 3rd axis



OECD field of science
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Engineering and Technology
Humanities
Medical and Health Sciences
Natural Sciences
Social Sciences
First author publications
Total publications
Fractional pubs.
Total citations
Avg. citations
Num. organizations
Num. nat. moves
Num. intl moves
Num. coauthored pubs.
Num. national pubs.
Num. intl. publications
Avg. collaborations
0
5
10
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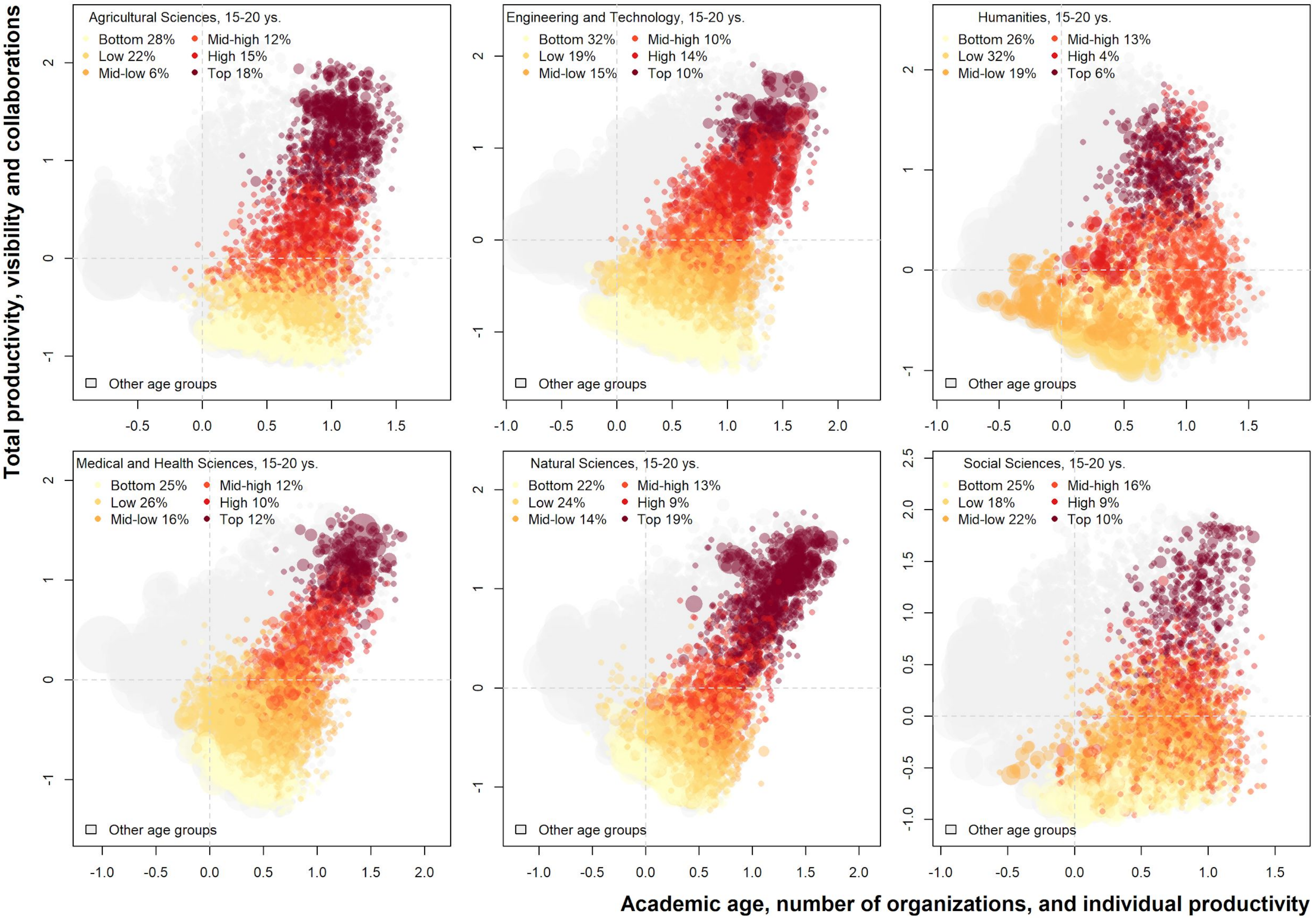
25
Percentage contribution to the 1st axis
Percentage contribution to the 2nd axis
Percentage contribution to the 3rd axis



15-20 years old

6, 8, 10 **clusters** were evaluated, **6** was chosen

We explored the hierarchical structure based on 12 variables.

How about collaboration networks?

Who co-authors publications with whom?

Do those in the top (most successful) collaborate only within their group?

Scopus co-authorship (bipartite / two-mode) network

Full network:

Vertices (nodes): **36,686,692**
Edges (ties): **93,102,217**

Giant component
(used for community detection):

Vertices (nodes): **34,768,821**
Edges (ties): **91,737,606**

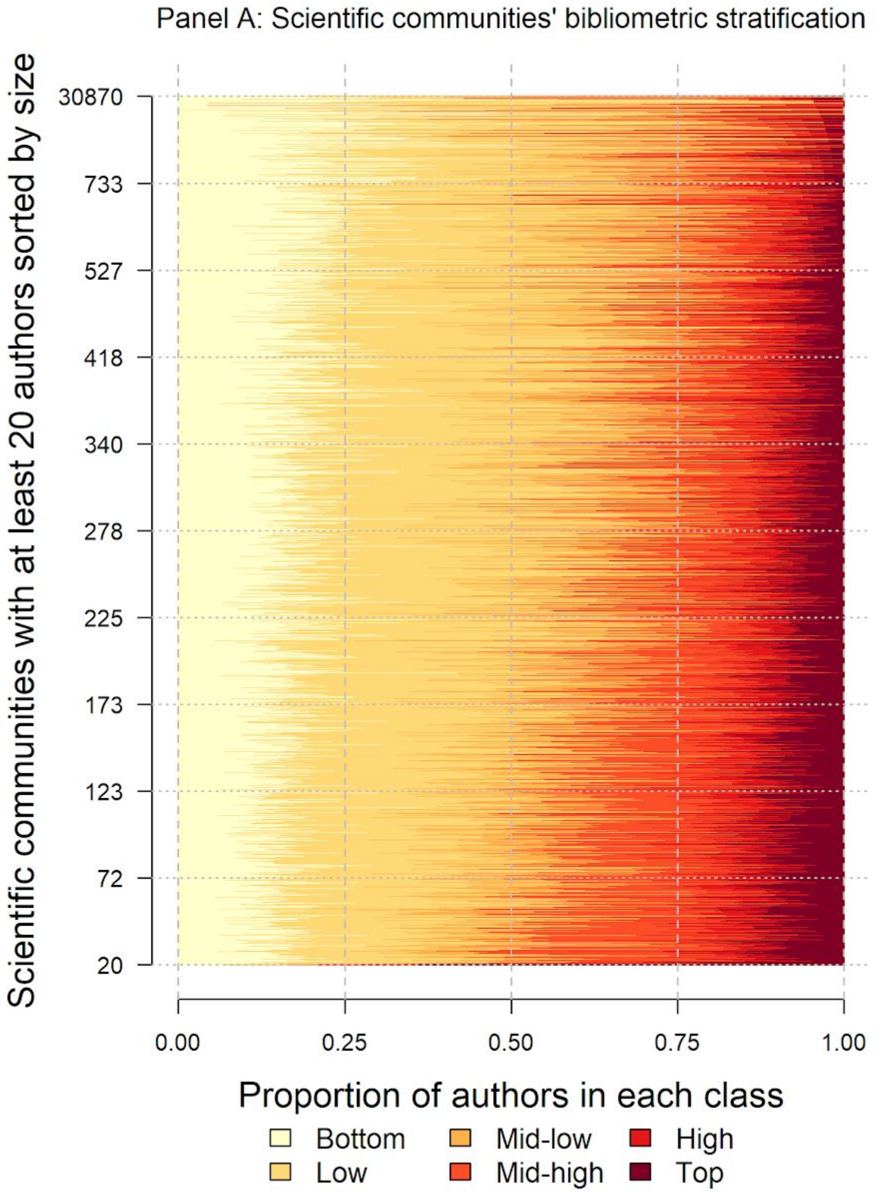
CD using Leidenalg in Python

with CPM bipartite

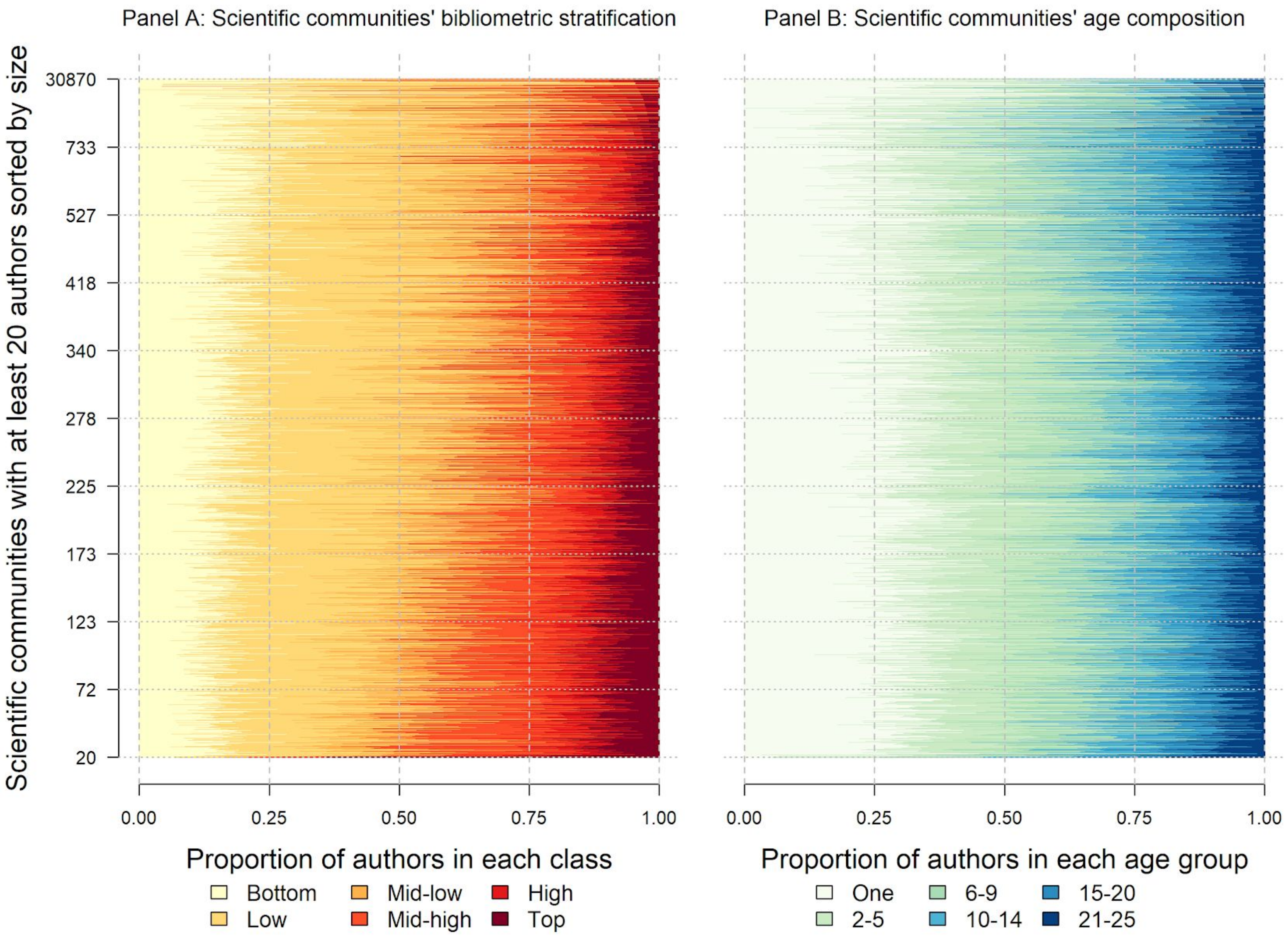
18 different Gamma for resolution parameter (e.g., G-comp's density)

Robustness of results with Networkit in Python, using default, parallel Louvain, parallel label propagation (parallel leiden did not run). Disproportionately larger share of specific **countries** or **disciplines**? No!

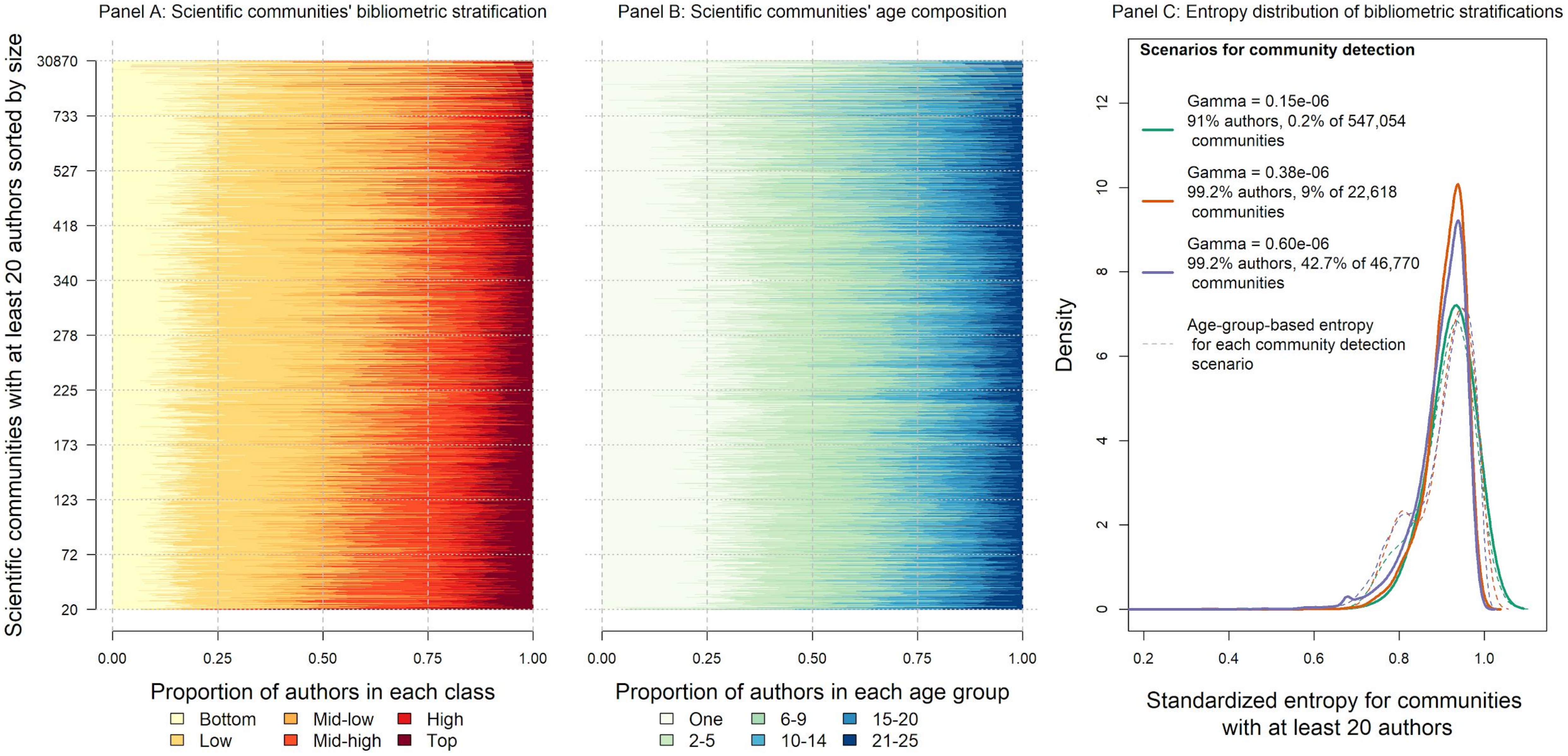
Infomap (igraph) and BimLPA (CDlib) failed/or kept running (past 2 weeks)

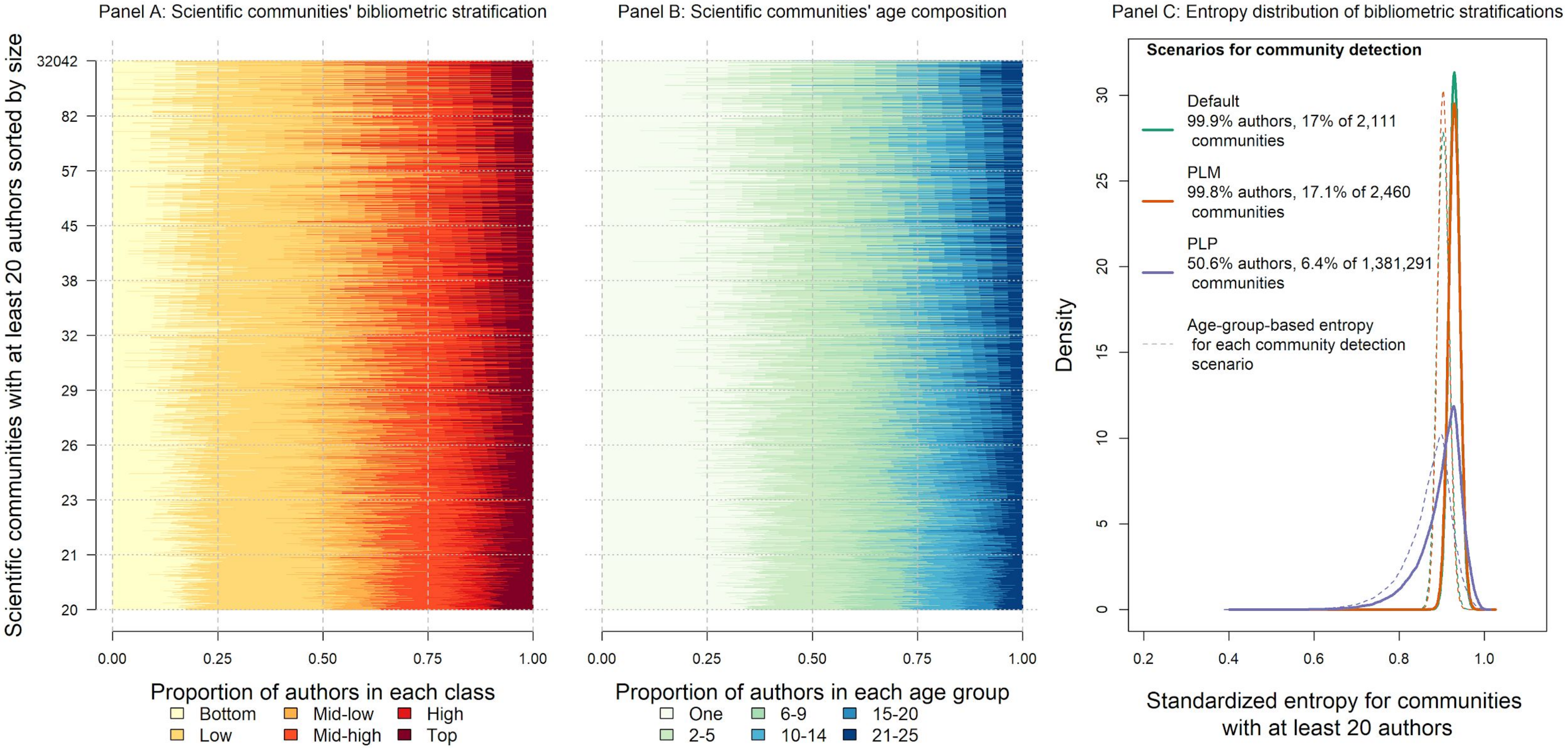


Panel A: Scientific communities' bibliometric stratification
Scientific communities with at least 20 authors sorted by size
30870
733
527
418
340
278
225
173
123
72
20
0.00
0.25
0.50
0.75
1.00
Proportion of authors in each class
Bottom
Low
Mid-low
Mid-high
High
Top

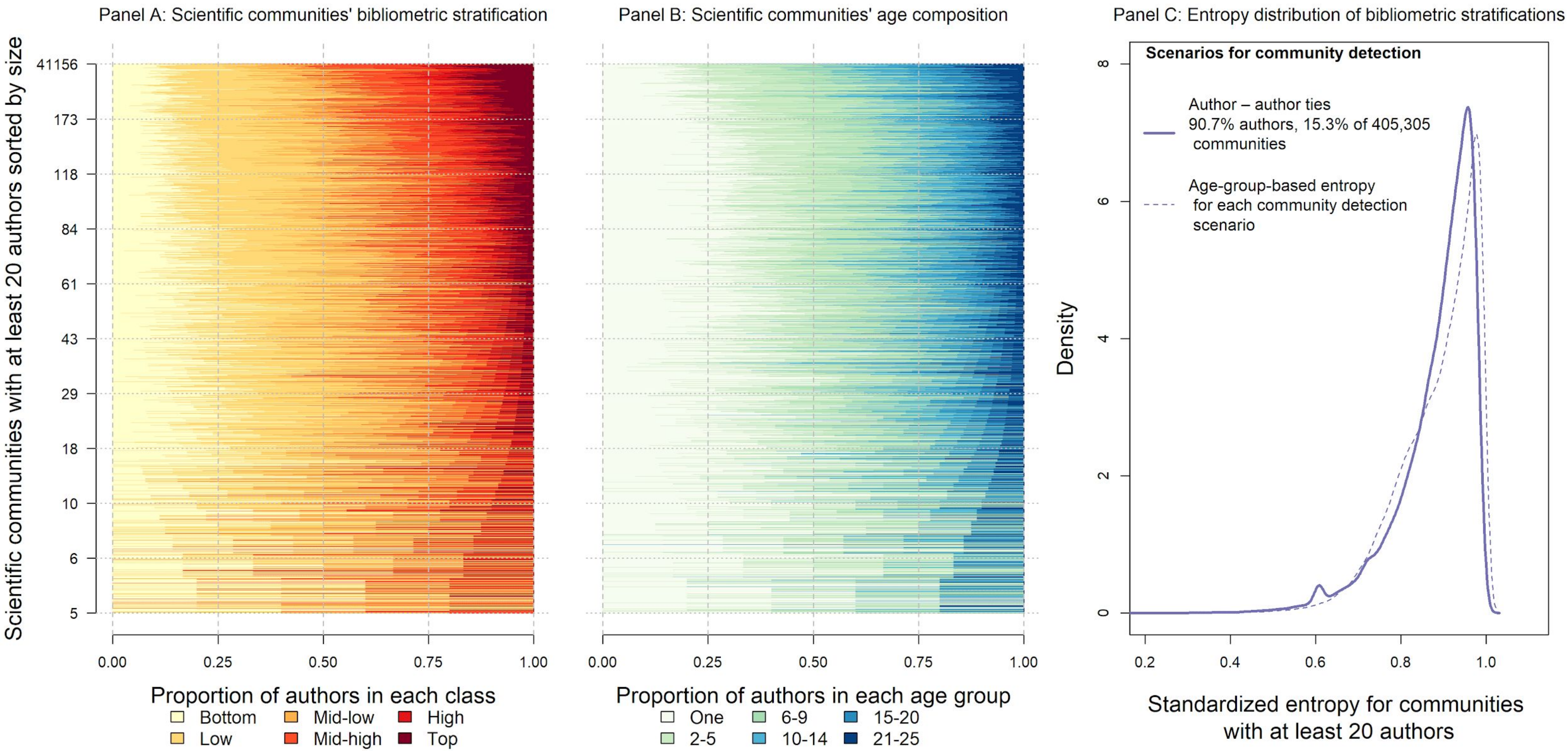


Panel A: Scientific communities' bibliometric stratification
Panel B: Scientific communities' age composition
Scientific communities with at least 20 authors sorted by size
30870
733
527
418
340
278
225
173
123
72
20
0.00
0.25
0.50
0.75
1.00
Proportion of authors in each class
Bottom
Low
Mid-low
Mid-high
High
Top
Proportion of authors in each age group
One
2-5
6-9
10-14
15-20
21-25

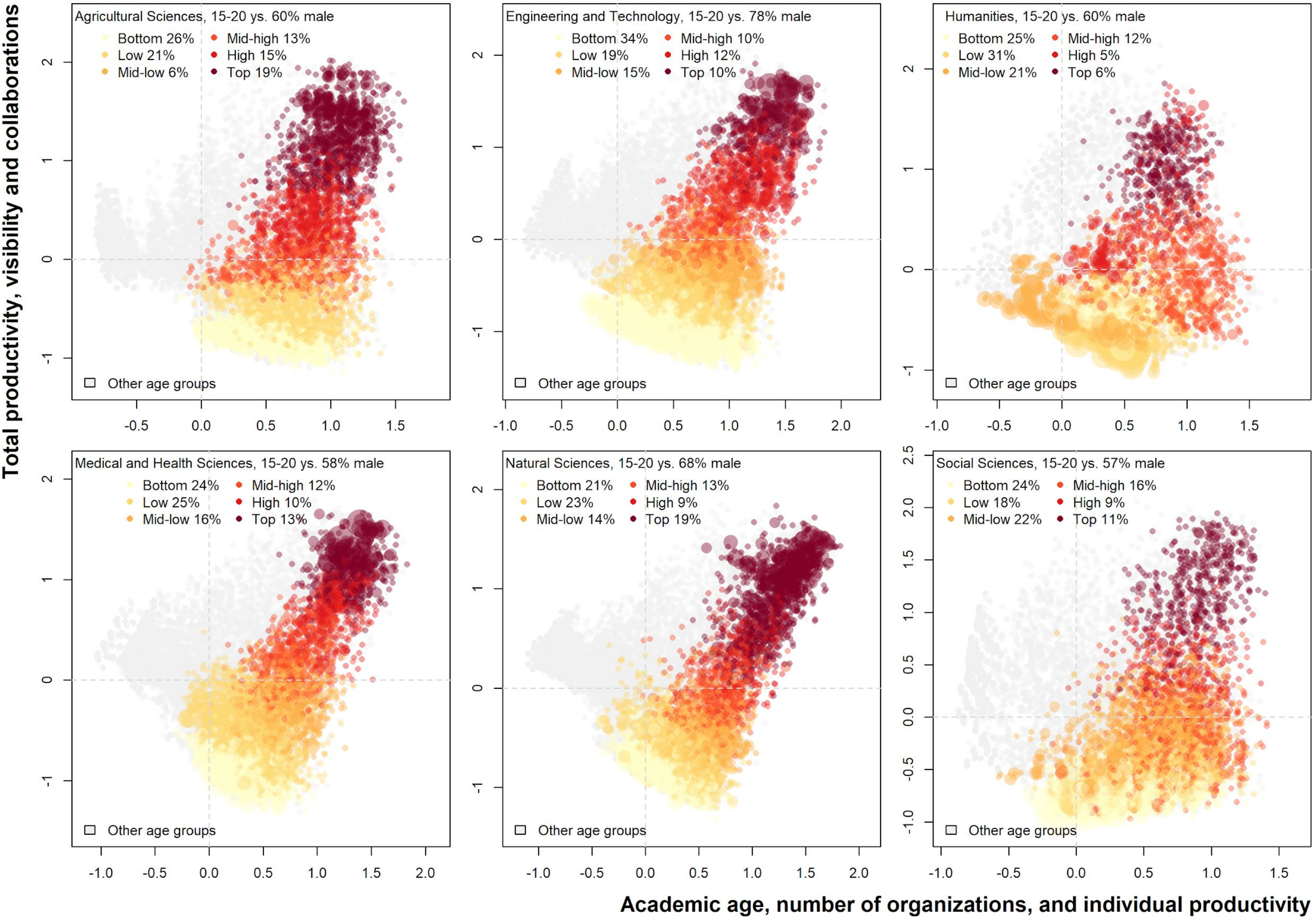

Panel A: Scientific communities' bibliometric stratification
Panel B: Scientific communities' age composition
Panel C: Entropy distribution of bibliometric stratifications
Scientific communities with at least 20 authors sorted by size
30870
733
527
418
340
278
225
173
123
72
20
0.00
0.25
0.50
0.75
1.00
Proportion of authors in each class
Bottom
Low
Mid-low
Mid-high
High
Top
Proportion of authors in each age group
One
2-5
6-9
10-14
15-20
21-25
Scenarios for community detection
Gamma = 0.15e-06
91% authors, 0.2% of 547,054 communities
Gamma = 0.38e-06
99.2% authors, 9% of 22,618 communities
Gamma = 0.60e-06
99.2% authors, 42.7% of 46,770 communities
Age-group-based entropy for each community detection scenario
Density
0
2
4
6
8
10
12
0.2
0.4
0.6
0.8
1.0
Standardized entropy for communities with at least 20 authors



Robustness analysis: Using Networkit, 3 CD algorithms, default, parallel Louvain, parallel label propagation

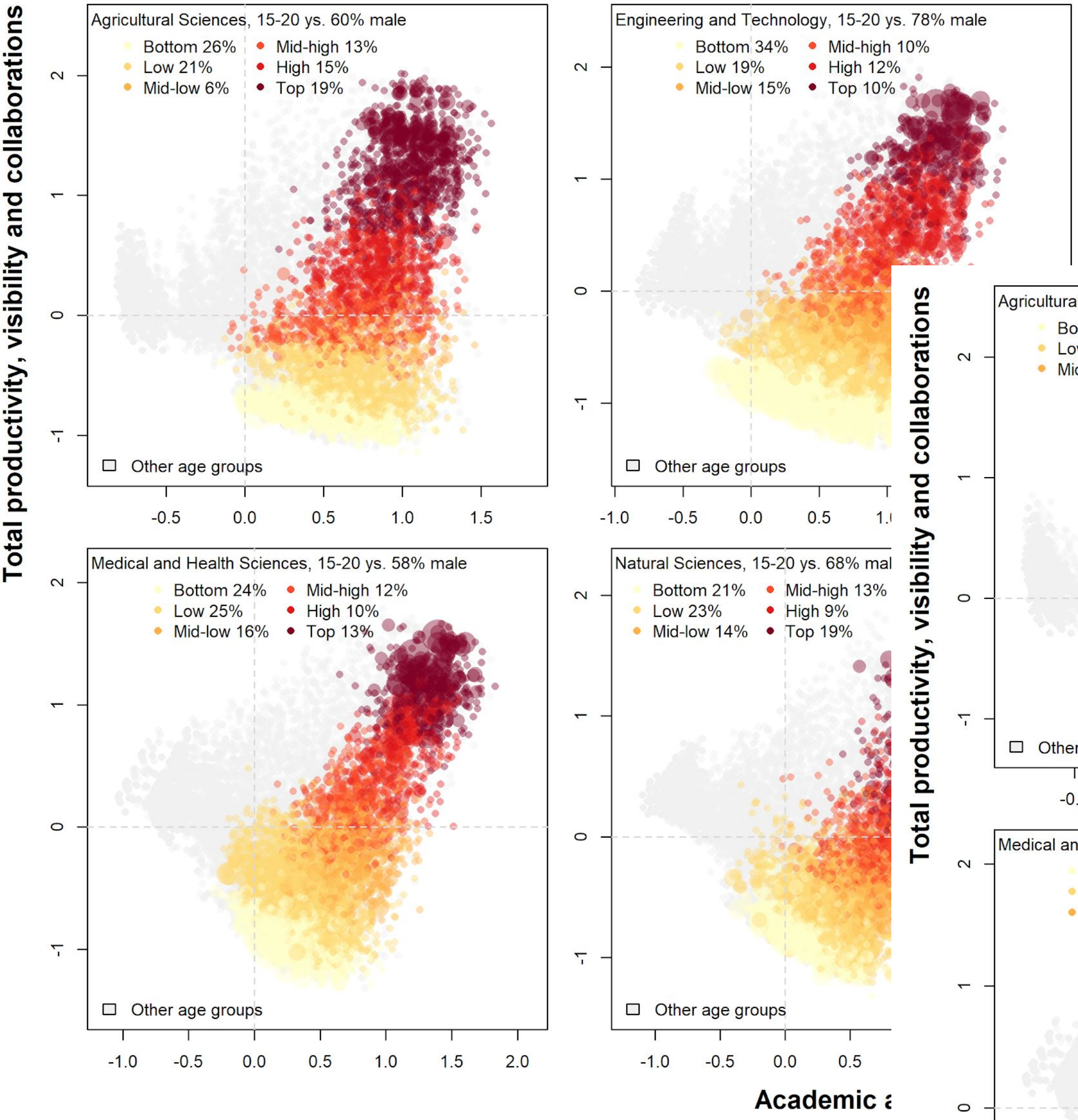


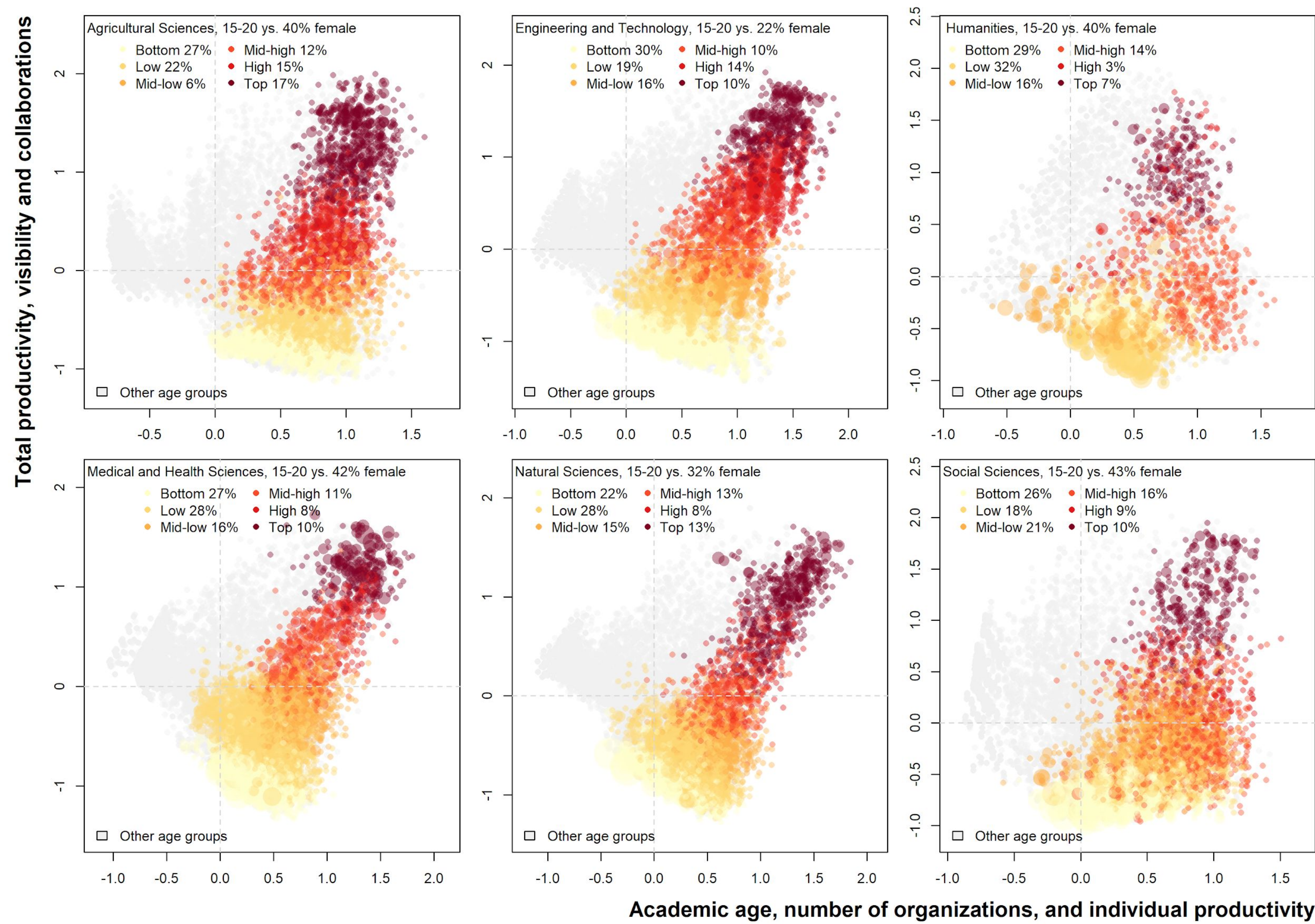
Robustness analysis: Using one mode projection of the network and Leiden algorithm (despite its cons)



How about gender's effect?

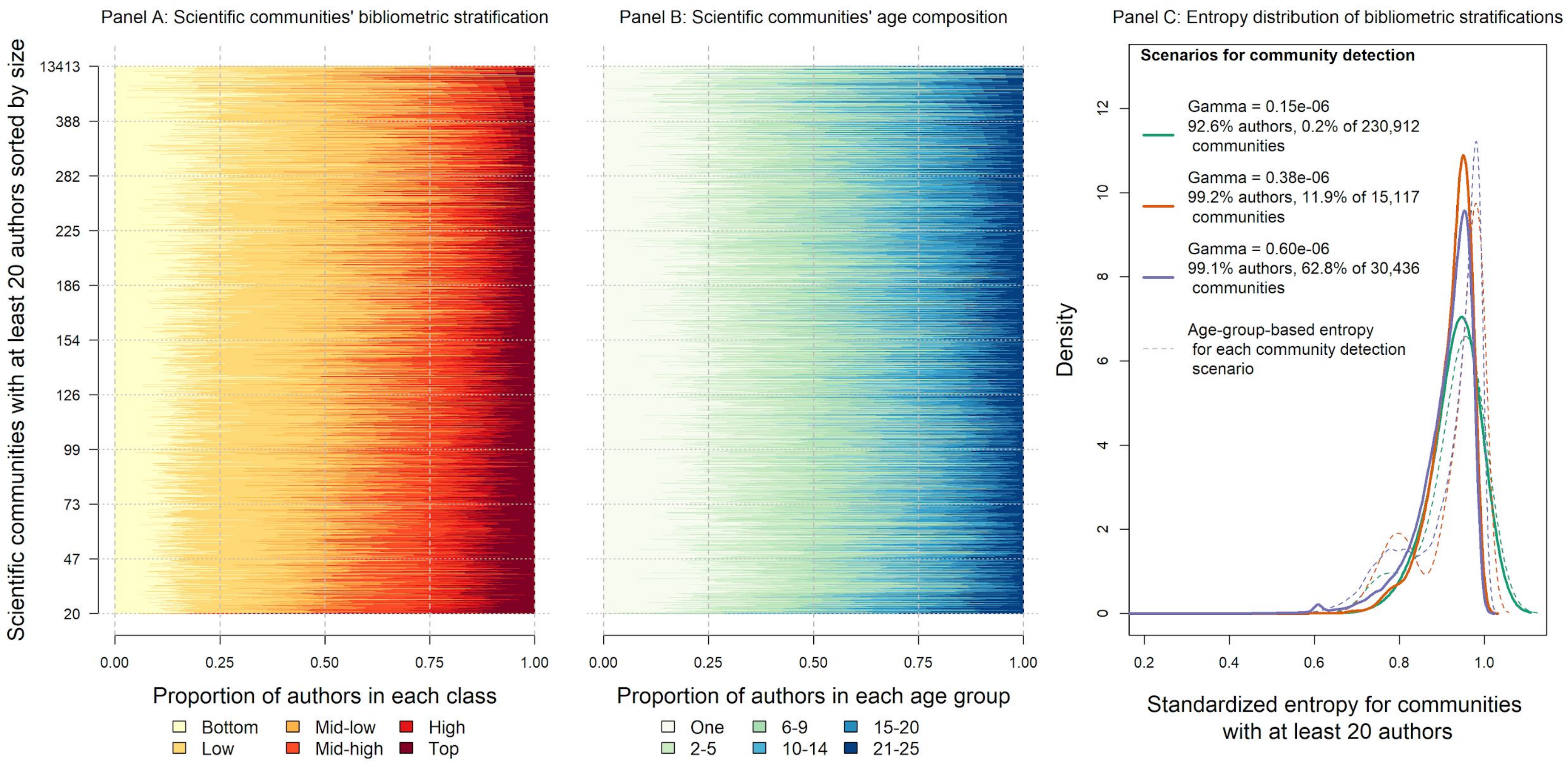
Men (left), Women (right)





How about gender's effect?

Men (left), Women (right)



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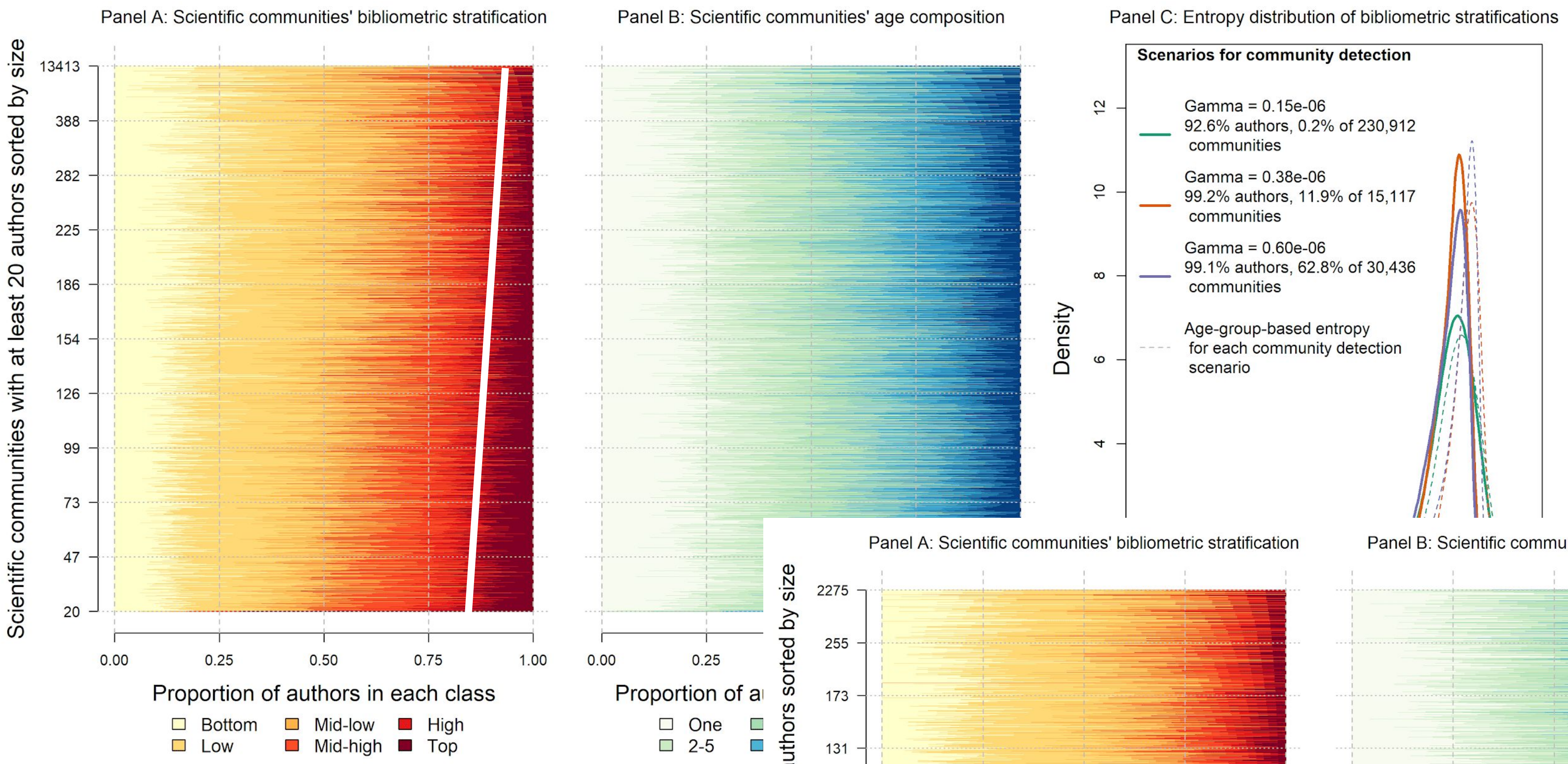
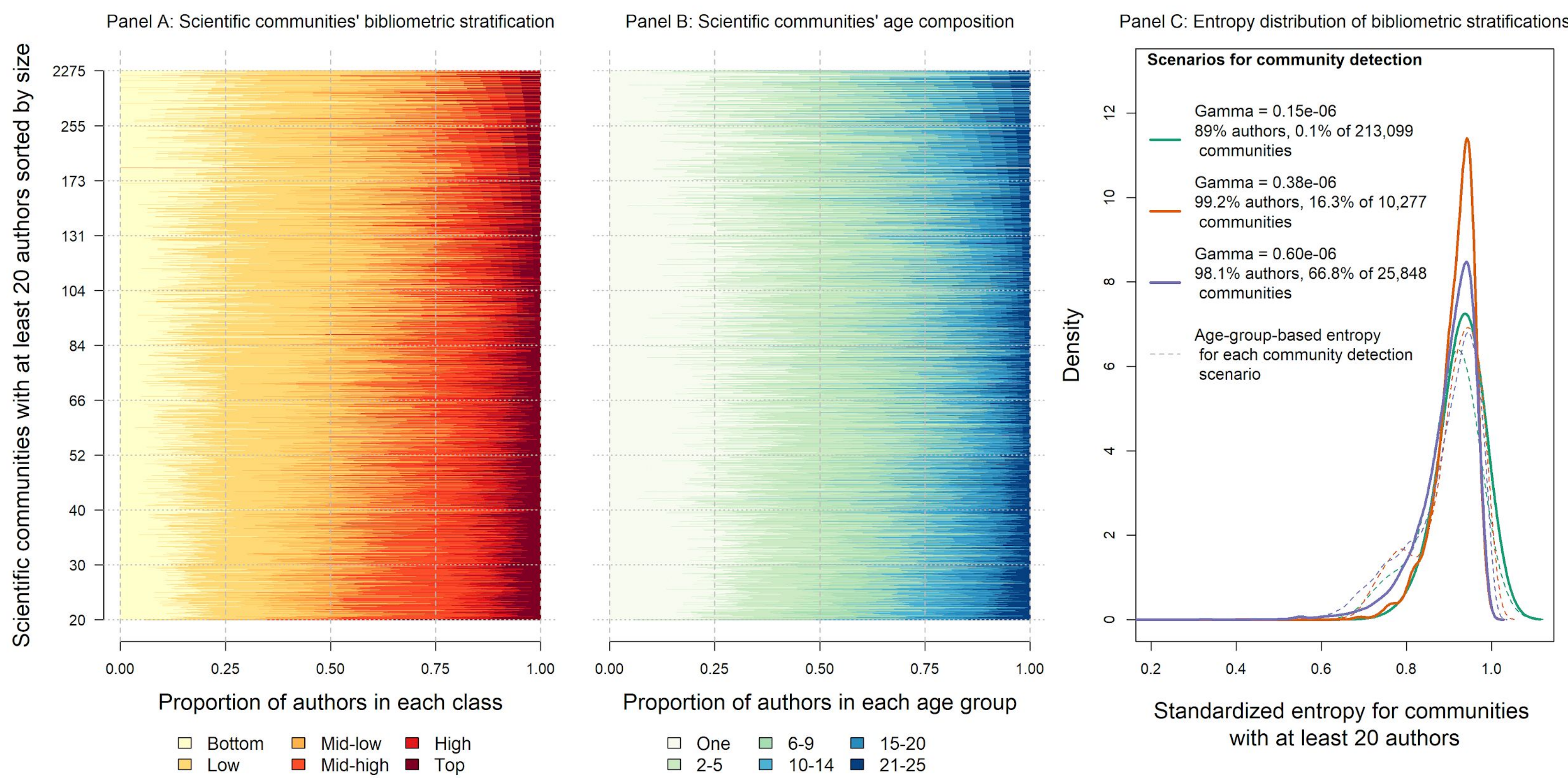
Men (left), Women (right)



How about gender's effect?

Men (left), Women (right)



Conclusions

- A **quantitative** assessment of the **global inequalities in science**,
- Using **bibliometric** data, across **fields of science** and **within** research communities.
- Claims on increased productivity, collaboration, and mobility are principally driven by a small fraction of influential scientists (top 10%). It is schockingly not the **gender** issue.
- A **stratified system** exists, and it is as strong as stratification by **academic age**. (a small top class, and large middle and bottom classes)
- Those at the top **succeed by collaborating** with a varying group of authors from **other classes** and **age** groups.
- Nevertheless, they are **benefiting disproportionately** to a much higher degree from this collaboration and its outcome in form of **impact** and **citations**.
- **Supervisor-supervisee** relationships inherently have an **age structure**, (Our observation goes beyond that (25% < 5 yrs)
- We propose considering academic age, career cohorts and composition of a multitude of bibliometric variables instead of solely relying on one-indicator explanations.

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Thank you for your attention!

Feedback, comments, and questions are very welcome.

Aliakbar Akbaritabar

Max Planck Institute for Demographic Research, Rostock, Germany

Akbaritabar@demogr.mpg.de

Andres F. Castro Torres

Center for Demographic Studies, Autonomous University of Barcelona, Barcelona, Spain

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Université de Montréal, Montréal, Canada.

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